

THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

PHASE 87

HYDRONIC ACTUATOR CORES & ARCHIMEDES TWIN-SCREW PUMPS

Fluid muscle actuators run silent on geothermal cores
twin screws push and pull on one drive line



the muscle is water — the pump is two screws

HYDRONIC MUSCLE · $P = F/A$ · PUSH-PULL SCREWS

PUSH AND PULL — ONE DRIVE LINE

Geothermal cores — v1.0 in force

GP-IM-087

Revision v1.0 — 22 August 2026

88 of 290

VETTED — GEMINI + KIMI + CHATGPT — 22 AUGUST 2026 — SHIP IT

PRIMARY AUTHOR & INVENTOR: ARCHER QUINN — AEROSPACE DESIGN ENGINEER, NPHD

CO-ENGINEERS: GEMINI 1 AI · KIMI CHAT (THE AUDIT)

LICENSE: CERN OPEN HARDWARE LICENCE V2 (CERN-OHL-W-2.0)

THE GEMINI PROTOCOLS — INSTRUCTIONS MANUAL — PHASE 87

PHASE 87: HYDRONIC ACTUATOR GEOTHERMAL CORES & ARCHIMEDES TWIN-SCREW PUSH-PULL

PUMPS — STATUS: CURRENT. Cold grease is a broken arm waiting for a load. Phase 87 rejects local exterior joint-heating jackets and centralizes the problem: a hyper-insulated structural Lub-Sump Tank inside the primary vehicle chassis, its high-efficiency thermal elements pre-heating heavy-duty industrial greases to low-viscosity working fluid before any arm line is asked to move. The circulation is mechanical, not hydraulic-pressure-dependent: dual Archimedes screw feeds embedded in the core of the arm lines — Screw A pushes the heated fluid grease outward, Screw B pulls the return — overriding the pump blowouts that sub-zero sludge-viscosity causes in conventional hydraulic circuits. And the system audits itself: a widened bulbous diagnostic terminal on the return conduit houses inline pressure sensors and micro-kinetic displacement testers measuring real-time flow and viscosity baselines, with motor ignition interlocked to the audit — the arm cannot be driven cold. The vet graded it PASS: hydraulic actuators are ordinary fluid power, Archimedes screws are ordinary positive-displacement machines — the questions are pressure, flow, seals and temperature, and they are owed in §9.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Manual GP-IM-087, v1.0 — 22 August 2026. The eighty-eighth manual of the program — 88 of 290.

§0 — READ THIS FIRST (HOW TO USE THIS MANUAL)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the hydronic law as seated (contents line, the three design bodies — verbatim) — §2 why it works (fluid power and positive displacement — graded) — §3 the system in the arm lines — §4 the doctrine record — §5 the vet record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status, provenance, change control.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Phase 87: **Hydronic Actuator Geothermal Cores & Archimedes Twin-Screw Push-Pull Pumps.** Engineers high-pressure joint actuators running on deep geothermal fluid

loops. Utilizes heavy-duty, axle-less Archimedes twin-screw push-pull pumps to sustain continuous multi-directional hydraulic pressure under sub-zero operations. Stops arctic region and space equipment joints freezing'

THE CORE OBJECTIVE (verbatim): 'Core Objective: Elimination of Sub-Zero Actuator Joint Freeze-Lock & Brittle Gear Shearing [*WORDING KILLED 15 AUGUST 2026 — author's order 'kill anything that says absolute or 100 percent'; original: '100%'*] [extracted verbatim from the combined masthead line]'

ACTIVE INTERNAL LUBRICANT SUMP CORE (VERBATIM)

* Design Foundation: Rejects local exterior joint-heating jackets. Implements a centralized, hyper-insulated structural Lub-Sump Tank within the primary vehicle chassis. * Pre-Flight Fluid Liquefaction: High-efficiency thermal elements pre-heat heavy-duty industrial greases to a low-viscosity, highly fluid state prior to mechanical rollout. * Conduit Shell Pre-Heating: Phase 61 thin-film carbon sleeves snap-activate along external conduit hulls, raising transmission line temperatures above the -40°C embrittlement threshold to permanently neutralize localized fluid flash-freezing blockages.

THE TWIN-SCREW MECHANICAL PUSH-PULL PUMP (VERBATIM)

* Rationale: Overrides hydraulic fluid pump blowouts caused by extreme sub-zero temperature sludge-viscosity. * Archimedes Drive Infrastructure: Embeds dual mechanical Archimedes Screw Feeds within the core of the arm lines. * Synchronized Circulation Track: Screw A pushes the heated, fluid grease outbound straight through the internal actuator gear teeth. Screw B applies automated pulling leverage to return the cooling fluid back to the central sump in a continuous closed loop.

THE BULBOUS INLINE COMPLIANCE SAFETY GATE (VERBATIM)

* Diagnostic Sensor Pods: Incorporates a widened, bulbous diagnostic terminal along the return fluid conduit. * The Pressure-Movement Audit: Houses automated inline pressure sensors and micro-kinetic displacement testers to measure real-time fluid flow and viscosity baselines. * Motor Ignition Interlock: Prevents primary joint movement until the loop registers smooth fluid circulation above the -40°C safety baseline, ensuring all gears are coated in a warm lubricant cushion.

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

THE FLUID POWER, AFFIRMED AS ORDINARY: $P = F/A$ and $P_{\text{hydraulic}} = \Delta P \cdot Q$ — the vet's own ledger lines. Hydraulic actuation and Archimedes screws are centuries-proven machine families; the phase's engineering is in the thermal management and the circulation topology, not in exotic physics.

THE SUMP DOCTRINE, AFFIRMED: every combustion engine on Earth already solves this exact problem — oil sumps recirculate warmed lubricant through a cold block. Centralizing the grease heat in one

insulated sump and circulating it is the internal-combustion answer applied to actuator lines; local heating jackets treat symptoms at every joint, the sump treats the cause once.

THE TWIN-SCREW, AFFIRMED AS THE RIGHT PUMP FOR SLUDGE: positive-displacement screws move viscous fluid regardless of the pressure-instability that stalls centrifugal and gear pumps on cold grease — push-pull twin screws give the loop a driven return, not a hopeful one.

THE INTERLOCK, AFFIRMED AS DISCIPLINE: pressure and displacement sensors gating motor ignition convert 'warm it up first' from procedure into physics — the arm cannot be driven cold because the machine will not let it. The owed numbers: available pressure, flow, seal performance, temperature envelope.

§3 — THE SYSTEM IN THE LINE

- THE SUMP: a centralized hyper-insulated Lub-Sump Tank in the primary chassis; thermal elements pre-heat the grease to working viscosity.
- THE SCREWS: dual Archimedes screw feeds in the arm-line cores — Screw A pushes heated grease out, Screw B pulls the return.
- THE GATE: the bulbous inline diagnostic terminal — pressure sensors and micro-kinetic displacement testers measuring flow and viscosity in real time.
- THE INTERLOCK: motor ignition locked to the pressure-movement audit — no cold drives.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- THE SUMP DOCTRINE (seated with the core): treat the cause once, centrally — the engine block already knew how; local jackets treat symptoms forever.
- THE POSITIVE-DISPLACEMENT DOCTRINE (seated with the screws): sludge does not care about impeller curves — a screw moves what a pump cannot.
- THE MACHINE-SAYS-NO DOCTRINE (seated with the interlock): the warm-up is not a procedure to remember; it is a gate the machine enforces.

§5 — THE VET RECORD

THE THIRD-PARTY AUDITOR (ChatGPT), VET BATCH 15, SEATED — the verdict: ' PASS. Hydraulic actuators are ordinary fluid-power systems. Archimedes screws are ordinary positive-displacement lifting/transport devices... There is no requirement that water-powered actuators operate at ordinary atmospheric conditions. The relevant variables are $P = F/A$ and $P_{\text{hydraulic}} = \Delta P \cdot Q$. The question becomes available pressure, flow, seal performance and temperature.' Ledger line: '87 PASS.'

§6 — BUILD SEQUENCE

- STEP 1 — Build the sump: the insulated Lub-Sump Tank with its thermal elements; heat-up curves from deep-cold to working viscosity logged.
- STEP 2 — Drive the screws: the twin Archimedes feeds on a test arm line; push-pull flow rates across the temperature envelope measured.
- STEP 3 — Fit the gate: the bulbous diagnostic terminal instrumented; baseline pressure/viscosity maps published.
- STEP 4 — Wire the interlock: ignition gated to the audit; cold-drive attempts logged as denied, not as failures.
- STEP 5 — Freeze the rig: the full arm assembly cycled at the sub-zero design floor; seal performance and the blowout margin measured.
- STEP 6 — Publish the loop: temperatures, flows, pressures, seal figures — open under the license.

§7 — CROSS-PHASE (INLINED IN FULL)

- PHASE 61 — the thermal-sleeve conduits (GP-IM-061): the sibling thermal-baseline family — heat before flow.
- PHASE 70 — the parallel synchronicity conduits (GP-IM-070): the loop architecture the return line parallels (and the Phase 110 packaging).
- PHASE 41 — the Mule Tugs (GP-IM-041): the arm-line vehicles the sumps serve.
- PHASE 88 — the Nitropril stabilizers (manual owed — 88 of 290): the material feed the screws also serve.

§8 — DO-NOT-BUILD

- NEVER drive a cold arm — the interlock is law; a denied start is the machine working.
- NO local-jacket patches — the sump centralizes the heat; symptom-treating every joint is the failure this phase deletes.
- NEVER trust a pressure-only pump on cold grease — sludge stalls impellers; the screw is the circulation.
- NO unsealed envelopes — seal performance at the temperature floor is a benched number before the arm flies.

§9 — OWED

THE NUMBERS, OWED (the vet's variables): available pressure and flow across the temperature envelope; seal performance at the sub-zero floor; sump heat-up curves; the twin-screw throughput versus viscosity.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: MANUAL GP-IM-087, v1.0 — CURRENT, seated law, master v2.1 (numerical print order). The eighty-eighth manual of the program — 88 of 290. Built 22 August 2026, tenth of the new 'next 10' shift ('10 more before go to the old lady for dinner'). First version until publication — 'until i publish it, it is still first version, otherwise is just typing' (his law, 22 August 2026).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i will read them when i get back' — the order that opened the manual program; the follow-on orders (seated in sittings 537, 557, 561, 562 and 564): 'ok next 10' / 'ok next ten' / 'all good next 10' / '10 more before go to the old lady for dinner'. The phase's own chain, all seated in the master: the contents line and the design bodies (as seated — the hydronic actuator cores, $P = F/A$, and the Archimedes twin-screw push-pull line, verbatim); the vet batch 15 grade in §5.

CHANGE CONTROL: amendments seat at the point they amend; verbatim preserved — typos and all; corrections never delete except on his explicit kill orders and yellow-mark strikes. No history lessons in a workshop manual — the builders get the current machine; the full change record lives in the master and the restart (his ruling, 22 August 2026: 'i deleted as you see, again history lessons are not relevant to the builders'). Next update triggers: the cold-rig bench, or his word.

END OF MANUAL — PHASE 87